RealQM Molecule Simulator: Validation on Small Molecules from Hydrogen and the Second Row

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Abstract

We describe a simulator for small molecules built from hydrogen, carbon, nitrogen and oxygen in RealQM, a real-space model in which each electron is a unit-charge density occupying a non-overlapping spatial domain, and the validation it is built to support. The model is given three valence charges—$Z_x = 1, 3, 4$ for hydrogen, carbon, and nitrogen with oxygen—and each molecule's observed geometry; it returns the binding energy and the geometry it settles at. Every other quantity is a solver setting, and the settings are declared once and never adjusted per molecule. That is a weaker claim than parameter-free and a more useful one: the construction has settings, which we tabulate, but no per-system adjustment, so each molecular number is a prediction rather than a fit. The central such declaration is the grid: one spacing, $h = 0.1\,a_0$, for every molecule in the list, treated as a commitment about what the model is rather than as a convergence parameter. At fixed h the grid size no longer carries resolution—it fixes the box—and the box must be fixed too, because mixed boxes were measured to corrupt a cross-molecule energy difference by a factor of 2.7. The declaration costs something and buys something. It costs the claim of being parameter-free: h becomes a constant of the model, fixed by measurement rather than derived, and we say so plainly rather than presenting the grid as a numerical setting. It buys falsifiability, since with h free any disagreement can be deferred to convergence, whereas with h declared every bond length and every binding energy is a prediction carrying no remaining adjustment. On that footing the simulator reports both observables directly: binding energy against the measured well depth D_e, and bond length against R_e. We work hydrogen through in full as the foundational case. Its binding energy is $0.1666 \pm 0.0018\,E_h$, where the uncertainty is measured scatter over a factor of two in the initialisation and not an estimate. That value falls *between* the two experimental conventions—4.5% below the well depth D_e and 1.2% above the directly measured D_0—which differ by more than the model's own scatter, so the model is accurate to a few percent on either reading and the sign of its error is set by a convention it cannot resolve. The declared box is worth a further four points, more than the scatter, and runs the unflattering way: a smaller box binds more strongly, so enlarging it costs accuracy rather than buying it. Its bond is 52% too long; and these are not independent defects but one, since the well's curvature is 49% too soft and $\Delta R = F/k_{\text{model}}$ turns a given force residual into a proportionately larger displacement—a conversion that must use the model's own curvature, the measured force constant understating the bond error by the stiffness ratio. Its bond error is almost entirely discretisation rather than model: an axisymmetric reduction, exact for a two-domain system on an axis, converges the same bond to $1.368\,a_0$—short by 2.3% and flat to 0.18% under a 2.5-fold refinement—with the energy minimum and the force zero agreeing to 0.06%. So the refinement sequence crosses experiment on its way to

something shorter, and the spacing at which the error vanishes marks a cancellation rather than an agreement. The two quantities we validate on are the *geometry the model settles at* and the *binding energy*—a length and a depth, both in the units the model claims to predict. Forces at observed geometries, which an earlier version of this work made primary, are demoted to a check: a residual converts to a bond error only through the model's own curvature, and using the measured force constant instead—as we did—understates hydrogen's error by twenty-five points. What the force retains is narrower and real: the total residual measures electronic non-convergence, and its agreement with $-dE/dR$ (0.06% in an exact axisymmetric reduction) is what licenses using forces for dynamics. It is not a figure of merit, and an unbound hydrogen molecule scores as well on it as the best entries in the list.

1 Introduction

RealQM [1] replaces the many-electron wavefunction by N unit charge densities occupying non-overlapping spatial domains with free interfaces, interacting by Coulomb's law with each pair counted once. Nuclei enter as kernels; in molecular work a filled core is replaced by a *pseudo-kernel* of reduced charge Z_x carrying only the bonding valence. The model is therefore a force law: kernels move under the electrostatic force of the charge distribution, and a predicted geometry is wherever that force vanishes.

What goes in, and what comes out. The model is given *three valence charges*— $Z_x = 1, 3, 4$, covering hydrogen, carbon, and nitrogen with oxygen—and each molecule's *observed geometry*. Everything else is a solver setting, and the settings are declared once and never adjusted per molecule. That is the load-bearing claim of this paper, and it is weaker than "parameter-free" while being more useful: the construction plainly has settings, listed in Section 3, but it has no *per-system* adjustment, so every molecular number is a prediction rather than a fit. A reader can check this by inspection, which is why the settings are tabulated rather than described.

What comes out is the *binding energy* and the *geometry the model itself settles at*, with the *dipole* as a third quantity that doubles as an acceptance test—a symmetric molecule must give $\mu \equiv 0$, so any value there is pure error and rejects the state before its energy is read.

Two limits of that economy belong here rather than in a closing section. Three charges covering four elements is *degenerate*: nitrogen and oxygen both take $Z_x = 3$ and are bit-identical calculations, so the model cannot distinguish them and no result involving both speaks to their difference. And the geometry prediction presently covers four molecules rather than thirteen, because rescaling a bond is implemented for diatomics only and three-centre relaxations oscillate (earlier work in this programme).

Validating such a model is awkward by the usual routes. Scanning the energy over geometry requires every point of the scan to be converged, and a scan assembled from partially relaxed states returns a smooth but meaningless curve. Relaxing under the force requires the force to be trustworthy, which is the thing under test. Both routes also need many calculations per molecule, and both return an answer whether or not the underlying states deserve one.

The geometry the model settles at—primary. The separation and angles the model itself reaches, against the measured ones. This is the observable with the fewest defects: it is a *length*, in the units the model claims to predict; it needs no second quantity to interpret; and it is not symmetry-protected, where a force residual is—CO_2 can contract symmetrically with $\sum_a \mathbf{F}_a$ staying exactly zero while every bond is wrong. It must be taken from a *commensurate, two-sided* scan:

released from the experimental geometry the molecule expands and stops short, giving $1.796\,a_0$ for H_2 against a true minimum of 2.134, so one-sided relaxation flatters the model by some ten points.

The binding energy—primary. E_{bind} from (2) against the measured well depth, reported against both conventions because they differ by more than the model's own scatter. It requires atomic references computed with the same construction on the same grid, which is why Section 4 treats the atom before any molecule.

The force at the observed geometry—a check, not a figure of merit. At the measured equilibrium the true force vanishes, so the residual is model error with experiment supplying the reference, from a single calculation. We previously presented this as the primary observable and that was a mistake, for a reason H_2 makes quantitative. Converting a residual to a bond error needs a force constant, and *which* one decides the answer: $F(R_e) = +0.1383\,E_h/a_0$ gives +27% divided by the measured $k = 0.3692$ and +53% divided by the model's own $k = 0.1875$, against a true minimum at +52%. So the conversion is right only with the model's curvature—a second quantity, itself 49% in error, requiring its own three-point scan—and the measured constant *understates* the model's error by twenty-five points. Every ΔR in Table 3 was converted that way and is a lower bound.

What the force retains are two narrower uses, both real. The residual total force measures *electronic non-convergence*, since it equals the net force of the cloud on the nuclei and vanishes only for a stationary state. And agreement with $-dE/dR$ is what licenses using forces for dynamics at all: 0.06% in the axisymmetric reduction, 3.2% in the three-dimensional solver. Neither is a measure of how good the model is.

The two primary quantities are not independent, and reporting either alone invites a wrong generalisation: a model can have the right depth and the wrong position if its curvature is wrong, which is exactly what hydrogen does. Section 5 works both through for H_2, the one system whose atomic reference involves no core at all.

What is in scope, stated once. At one electron this model is the simplest Coulomb model there is, and it is not distinguishable from the standard one: minimising $\frac{1}{2}\int |\nabla u|^2 - \int (Z/r)\,u^2$ subject to $\int u^2 = 1$ is the Schrödinger variational problem, with the same functional and the same minimiser, so $E(\mathrm{H}) = -0.5$ follows by *reduction* and not by any fitting. Hydrogen accordingly tests the *discretisation* and not the model.

One calibration, on hydrogen, and then none. The discrete realisation does need one constant that the continuum model does not: a grid cannot represent $1/r$, so the kernel carries a softening σh. We fix it once, by requiring

$$E(\mathrm{H}) = -0.500000 \qquad \Longrightarrow \qquad \sigma = 0.627 \;\text{ at } h = 0.1\,a_0, \tag{1}$$

measured on a four-point ladder (-0.508456, -0.500962, -0.493833, -0.486591 at $\sigma = 0.4, 0.6, 0.8, 1.0$, linear in σ to better than a percent). Verified at the declared grid rather than extrapolated to it: $\sigma = 0.627$ returns $E(\mathrm{H}) = -0.500067$ at box $16\,a_0$, an error of 0.013% (and -0.499901 at box $20\,a_0$), confirming also that the ladder's value carries across boxes as the near box-independence of a single atom requires. This is the *only* calibration in the paper, and three things about it matter. It is fixed on a *one-electron* system, where the model has no content of its own and the reference is exact, so no molecular observable is consumed. It is a single number, not a function. And it is fixed before any molecule is computed.

Everything molecular then follows with *no further calibration*: the same σ, the non-overlapping densities, the interface condition, and the observed geometries. There is nothing left to adjust, per molecule or otherwise, so every entry in the tables below is a prediction. That is the sense in which this is the simplest model of a molecule that can be built on Coulomb's law: the one-electron problem is the minimal Coulomb model, the only thing added is that the densities do not overlap and must meet somewhere, and the only thing fitted is the one constant that repairs the grid.

We distinguish this from the *declared settings*—the spacing h, the box, and the five coefficients of the free-boundary update (6). Those are numerical choices, listed in Section 3 and not fitted to any observable; they are declared rather than calibrated, and none of them is adjusted per system. One calibration and a declared numerical scheme is a weaker claim than "parameter-free" and a more useful one, because it is checkable: there is exactly one number in this paper that was set by looking at an answer, and it was not a molecular answer.

It is worth putting this the other way round, because it is the stronger statement. Schrödinger's equation for hydrogen *is* the minimal continuum model that can be built from a Coulomb attraction: a kinetic penalty quadratic in ∇u is the lowest-order rotation-invariant term that prevents collapse, and no simpler functional binds at all—a purely local kinetic density, for instance, does not. Hydrogen's exactness is therefore a property of those two ingredients and not of either theory's account of *many* electrons. The standard formulation and ours contain exactly the same ingredients at one electron and are both exactly right, so hydrogen discriminates between no formulations whatsoever. Nothing in this paper rests on it, and we would regard any programme that advertised one-electron agreement as support for a many-electron construction as having claimed the ingredients' success for its own.

The model acquires content only at two electrons, where the partition, the interface condition and and the question of what keeps two electrons apart all appear for the first time—so that is where its first genuine test is, and no agreement at one electron is evidence about molecules.

No molecule should be expected to come out exactly, H_2 included. This follows from the two paragraphs above rather than from any particular disappointment. At one electron the construction coincides with the standard minimal model; at two it both *adds* a partition and answers the question of inter-electron repulsion *differently*, and those changes are structural rather than perturbative. It is therefore a different theory precisely where molecules begin, and exact agreement with the other theory's answers would be the thing requiring an explanation. We say this in advance because it determines what the tables below can mean: a 20% energy error is the expected size of a known omission, not a numerical failing to be driven down, and the useful question is whether the error is *systematic*—one sign, one scale, cancelling in differences at fixed composition—which is a testable claim and the one we make.

A related point of hygiene. The value $-1.174470\,E_h$ that H_2's total energy is conventionally measured against is a Kolos–Wolniewicz *computation*, not a datum, and a computation in the very theory whose many-electron content this construction declines; so comparing a total energy to it sets one theory against another rather than against nature. The total electronic energy is in any case not an observable. H_2's observables are D_0, R_e, and a dipole that must vanish by symmetry, and those are what Section 5 reports. From two electrons onward two distinct mechanisms operate and should not be run together. The partition makes overlap *impossible*—a constraint, not a penalty, since the Coulomb integral is finite even for coincident smooth densities and so could never forbid coincidence by itself. The Coulomb term instead resists *compression*, rising as two domains are pushed together. And the partition generates *nothing* between equivalent domains across a mirror plane, where the masked and continuous constructions agree to 4×10^{-8}. We state

this as a measured property of this construction rather than as the absence of a term from another one: a joint antisymmetric wavefunction is not an observable, and a theory built on distinguishable densities on disjoint domains has no such object to antisymmetrise, so it is not *omitting* anything—it is answering the same question with different machinery, and the answer is testable on its own. The consequence is not a scatter but a signed bias: the model over-binds, by tens of percent, wherever two equivalent units face each other. On H_2 a strict-interface construction with hydrogen calibrated exactly gives E_{bind} too large by $+59.6\%$ at its own minimum while placing the bond within -3.7%; on helium, with the same softening, -3.027 against -2.9037.

We therefore do not claim absolute binding energies, and the reader should not read the E_{bind} column as one. What the construction does deliver is *geometry*, and *differences at fixed composition*, where the common atomic error cancels. Those are the two claims this paper makes. A one-parameter interface energy that buys the binding energy is available and we have measured it—it takes H_2 from $+59.6\%$ to $+18\%$—but it costs the bond, moving it from -3.7% to $+7.6\%$, and it costs the description of the model as parameter-free. We report the unmodified construction instead, and state its energy error rather than absorbing it.

2 Model and method

Throughout, the binding energy is

$$E_{\mathrm{bind}} \;=\; \sum_i E(\mathrm{atom}_i) \;-\; E(\mathrm{molecule}), \tag{2}$$

with every atomic reference computed in the same construction, on the same grid, at the same softening as the molecule—a requirement rather than a convenience, since a reference taken at different settings turns a model error into a settings error.

Each electron m carries a density u_m^2 supported on a domain fixed by a weight w_m, with $\int w_m u_m^2 = n_m$ the domain's occupation. Domains compete for territory, so the partition has free internal boundaries. The energy is

$$E = \sum_m \Big[\tfrac{1}{2}\int w_m |\nabla u_m|^2 - \int w_m K u_m^2\Big] + E_{ee} + \sum_{a<b} \frac{Z_a Z_b}{R_{ab}}, \qquad K(\mathbf{r}) = \sum_a \frac{Z_a}{r_{\mathrm{soft},a}}, \tag{3}$$

with every term weighted by w, electron–electron repulsion counting each pair once by the $i \neq j$ rule, and $r_{\mathrm{soft}}^2 = |\mathbf{r} - \mathbf{R}_a|^2 + (\sigma h)^2$ a kernel softening on the grid of spacing h. Calculations use bare kernels ($r_c = 0$) throughout, the declared grid of Section 3, and relaxation to a fixed step count with early termination disabled—the latter because the built-in stopping criterion compares energies between wall-clock samples and was found to report convergence on a plateau from which the state was still moving.

2.1 How the forces are computed

The force on a nucleus is the classical electrostatic force of the charge distribution, the content of the electrostatic force theorem [4, 5]; in a model built from charge densities it is definitional rather than derived. It is obtained by differentiating the discretised energy analytically rather than by sampling its gradient. From $V_{eK} = -\sum_a Z_a \int \rho / r_{\mathrm{soft},a}$,

$$\mathbf{F}_a \;=\; -\frac{\partial V_{eK}}{\partial \mathbf{R}_a} \;=\; Z_a \int \rho(\mathbf{r})\, \frac{\mathbf{r} - \mathbf{R}_a}{r_{\mathrm{soft},a}^3}\, dV, \tag{4}$$

a volume integral over the grid. It carries no averaging radius and is never evaluated at the nuclear cusp, and because it differentiates the same softened kernel from which the energy is built it agrees with $-dE/dR$ by construction, up to the term arising from the partition boundary moving with the nuclei—the analogue here of the basis-set derivative term of Pulay [6].

The alternative of sampling $Z_a \nabla V_H$ at the nucleus and averaging over a sphere of radius a to suppress grid noise is not a well-defined quantity on this mesh: holding H_2 fixed at $R = 1.68\,a_0$ and varying only a from 0.24 to $0.96\,a_0$ carries the force from -0.050 to $+0.027\,E_h/a_0$, with no plateau and a change of sign, since the averaging bias is $O(a^2 \nabla\rho)$ and $\nabla\rho$ is largest exactly at the cusp. Equation (4) is used throughout; its agreement with $-dE/dR$ was measured on H_2, where it predicts the model's own minimum to 0.2%.

3 One mesh for every molecule

All calculations reported here use the GPU solver `mol_fast` on a single declared grid:

$$h = 0.1\,a_0, \qquad 160^3 \text{ cells}, \qquad \text{box } 16\,a_0, \qquad \sigma = 0.627 \;\; \text{(calibrated, (1))}, \tag{5}$$

the same for every molecule in the list. The grid is not adjusted per system and no mesh ladder is run to justify a number. This requires defending, because it inverts the usual practice.

At fixed h, the grid size is the box. Since box $= nh$, declaring h turns the choice of grid size into a choice of how much vacuum surrounds the molecule, not of how finely it is resolved. The bond length is nearly insensitive to that: holding $h = 0.1$ and enlarging the box from 10 to $14\,a_0$ moves H_2 from 1.8855 to $1.8921\,a_0$, a change of 0.35%.

Why $16\,a_0$ and not more. Two constraints meet here and the second is the binding one. The box must be large enough to hold the polyatomic entries with room for their electron clouds, which rules out $12\,a_0$. And it must be small enough to compute: at $h = 0.1$, box $20\,a_0$ is 200^3 cells, and two-electron runs at that size exhaust the browser's resources and return no energy at all, while 160^3 runs to convergence in minutes. So $16\,a_0$ is not a compromise between accuracy and convenience—it is the largest box at which these calculations exist. We prefer to say that plainly rather than to declare a grid the computations cannot reach.

The declared box costs accuracy on the acceptance test, and we report it. Enlarging the box is not free in the direction one would hope. At box 20, H_2 at $R = 1.9\,a_0$ returns a dipole of 0.067 D where reflection symmetry requires identically zero, against 0.006 D for the same configuration at box 12—eleven times worse, with 0.097 against 0.010 at $R = 2.1$. The free boundary has more exterior to drift into and nothing anchors it there. Since a symmetric molecule's dipole is a pure error measurement needing no reference, the declared grid is measurably worse on the one test that cannot be argued with, and the reader should weigh the dipole column accordingly.

Where $h = 0.1$ comes from. Not from a physical argument. It is the finest spacing at which these systems compute in a common web browser—160^3 cells at the declared box—and the simulator is deliberately a browser program so that any reader can reproduce every number here without installing anything. That is an accessibility choice with a price, and the price is resolution: **the precision to expect from this table is fair rather than high**, several of the errors are mesh-limited rather than model-limited, and a finer mesh is the obvious next step that the constraint forbids. Where an h-ladder exists we report it rather than a single value.

The box must be fixed too. This is the reason the grid cannot follow the molecule. Comparing two systems computed in differently sized boxes inflates their energy difference: CO gave $0.0369\,E_h$ on mixed boxes against $0.0137\,E_h$ on matched ones, a factor of 2.7. A table of binding energies assembled from per-molecule boxes is therefore not a table of model errors but of box errors, and the only remedy is one box for the whole table, sized by its largest member.

What the declaration costs. It costs the description of the model as parameter-free. With h declared, h is a constant of the model, fixed by measurement and not derived from anything; and (5) is not an innocent choice, since $h = 0.1$ is the spacing at which the binding energy of hydrogen agrees with experiment while its bond is 52% too long (Section 5)—and the bond error there is discretisation, the converged value being short by only a few percent. A declared mesh is therefore not neutral between observables: this one is chosen where the energy is right and the geometry is not. A genuine constant of nature would be pinned by something physical; the natural candidate here is the kernel softening length $\varepsilon = h$, and at $0.1\,a_0$ that exceeds a real nuclear radius by four orders of magnitude, so no physical length currently fixes it.

And h is not the only one. The free boundary is advanced by

$$w \;\leftarrow\; w \;+\; \lambda\,\delta t\,|c|\,\nabla^2 w \;+\; \mu\,\delta t\,c\,|\nabla w|, \qquad c \;=\; u_m - \max_{n\neq m} u_n, \tag{6}$$

clamped to $[0, 1]$, with the occupancy set to zero below a floor w_f wherever another domain clearly owns the cell. The fixed point $c = 0$ is the amplitude-continuity condition of Section 5.2 and is part of the model; the coefficients are not. Declaring the grid therefore means declaring, alongside h: the interface sharpness $\lambda = 0.05$, the front speed $\mu = 1$, the residue floor $w_f = 0$, the relaxation step $\delta t = \nu h^2$, the initial territory extent $R_{\text{init}} = 3\,a_0$—and the box itself, which Section 5.1 measures at four points of hydrogen's binding energy and which therefore cannot be presented as a container sized to fit the molecules. We list them because an unlisted regularisation is indistinguishable from a fitted one.

One declared quantity is not a parameter of the dynamics at all. The kinetic energy is a *reported* quantity, and on a diffuse occupancy there is more than one way to report it. The functional the evolution actually minimises weights each face by the average occupancy across it, $T = \frac{1}{2}\sum_{\text{faces}} w_{\text{face}}(\delta u)^2 h$, which is what the amplitude update performs gradient descent on; this is the form declared here. Two alternatives are equally available: a hard gate, counting the *full* unweighted gradient wherever w exceeds a threshold and nothing below it, and the kinetic energy of the assembled global field $\Psi = \sum_m w_m u_m$, which is the object the model's $\Psi \in H^1$ requirement actually names. Measured on H_2 at box 12, $h = 0.10$, $D = 2.1\,a_0$—one converged state, scored three ways, with the occupancies and amplitudes identical in all three:

scoring of T	T	E	E_{bind} vs D_e
hard gate	0.9103	−1.137236	−5.9%
face-averaged (declared)	0.9197	−1.127903	−11.1%
assembled Ψ	0.9286	−1.1190	−14.0%

Eight points of D_e separate the extremes, from bookkeeping alone. The ordering is uncomfortable: the *most* internally consistent scoring is the furthest from experiment, and the gated form that agrees best with it is the one that does not correspond to the functional being minimised. We declare the face-averaged form because it is the one the dynamics minimises, not because it is

the one that flatters the model, and we report the spread so that no reader mistakes a choice of accounting for a result.

There is a second reason to declare it, independent of the first. The assembled field $\Psi = \sum_m w_m u_m$ is the object the model's $\Psi \in H^1$ requirement actually names, so the scoring closest to T_Ψ is the one closest to the stated construction. Measured on N_2 at box 16, $h = 0.10$—one state, both scorings, $T_\Psi = 10.4302$ throughout:

scoring	T	E	$T - T_\Psi$
gated	10.3800	-14.718135	-0.0502
face-averaged	10.4261	-14.672033	$\mathbf{-0.0041}$

The face-averaged form is twelve times closer. The entire difference between the two is kinetic—$\Delta E = \Delta T = 0.046102$ to six figures, which is the check that the choice touches nothing else—so the two scorings differ by $0.046\,E_h$ on N_2's total while agreeing exactly on its density, its geometry and its dipole. Nothing in the geometry depends on it: the three differ only in E, and six gate values give bit-identical bonds.

Their measured sensitivity on H_2 is the reason this is a short list rather than a serious one. Reducing λ by a factor of four moves the bond by 0.4 points; doubling R_{init} moves it by 0.2 points and the energy by $0.006\,E_h$. The floor is the exception: raising it from 0 to 0.05 lengthens the bond by up to 2.9 points, monotonically and saturating, and—awkwardly—the setting that leaves no spurious residue behind a retreating front makes the bond *worse*, which is why $w_f = 0$ is declared rather than defaulted quietly. The front speed μ has not been laddered on H_2 at all; it was fixed on a one-electron test where $\mu = 2$ was stable and $\mu = 10$ failed to conserve charge, and that is the outstanding gap in this list.

What it buys. Falsifiability. With h free, any disagreement with experiment can be deferred to convergence, and no single calculation is ever answerable for its result. With h declared, every bond length and every binding energy in the table is a prediction carrying no remaining adjustment, and the whole table stands or falls together. The sharp test, which we cannot yet run, is whether *one* spacing fits both observables at once; Section 5 shows that hydrogen's two observables currently want different ones.

4 The atom: homogeneous inner shells, sector-split valence

The binding energy (2) is a difference against separated atoms, so the atomic model is part of the molecular model and is stated first.

An atom is a kernel carrying the *full* nuclear charge Z with every electron represented as its own charge density, arranged in radial shells. The inner shells are **homogeneous**: spherical, non-split, multi-occupancy domains, whose intra-shell repulsion the solver reduces exactly for a shared shape, retaining $(n-1)/n$ of the self-Hartree, which is the identity

$$\sum_{i<j} \iint \frac{\rho_i \rho_j}{r} \;=\; \frac{n(n-1)}{2} \cdot \frac{1}{n^2} \iint \frac{\rho\rho}{r} \;=\; \frac{n-1}{2n} \iint \frac{\rho\rho}{r}. \tag{7}$$

The outermost shell is **sector-split**: divided into angular domains rather than carried as one sphere. Oxygen is $2 + 3 + 3$ on a $+8$ kernel, the configuration obtained from atomic calculations independently of any molecular input [2].

Three properties of this construction decide what the molecular results can mean.

The core excludes the valence by occupying space, not by decree. There is no inner cutoff and no core radius. The valence is held out of the core region by the core electrons themselves, through the same non-overlap that separates any two domains—and that exclusion is *compressible*: a valence pressed inward can squeeze the core, where a radius would simply block it. This distinction is not cosmetic. A radius calibrated to reproduce an atom's valence energy comes out at $0.92\,a_0$ for carbon, three and a half times the $0.263\,a_0$ Slater estimate of its $1s$ radius, and two such spheres on a second-row diatomic leave only $0.50\,a_0$ of bonding region between them: the calculation then converges reproducibly to an asymmetric state. A core that yields transfers to molecules; a radius sized to fix the atom does not.

The sector split changes the sign of the dipole, and we do not know why. Measured on OH with everything else held: the same valence electrons give $\mu_x = +0.778$ (correct sign) as two angular domains and -0.470—inverted—as one. Same kernel, same electron count, same geometry; only the partition differs. The effect is large and reproducible, and it is the sharpest discriminator we have between constructions, because a sign cannot be rescued by cancellation.

We decline to explain it by saying a single domain cannot polarise, which would be wrong: a domain's density is a free field over its territory, constrained only by $\int wu^2 = n$, so it may perfectly well carry more charge on one side than the other. A domain explicitly *flagged spherical* is shape-locked and can only translate or resize, and that is the case the nested construction uses for its inner shells—but it does not apply to a free-form single domain, which is what the measurement above compares. What the split supplies is therefore not polarisability as such. The mechanism is open, and we report the measurement without it.

Hydrogen is the one atom with nothing approximated. One electron, one domain, no core and no shell structure, so its only error is discretisation, and the calibration (1) removes it. This is why H_2 is worked through first in Section 5: every error there belongs to the grid and the interface treatment, not to the atom. The calibration does not carry to a deeper kernel, and a one-electron ion shows it with nothing else involved: He^+ at $\sigma = 0.627$ gives -2.053249 against the exact -2, 2.7% too deep, with its kinetic energy 5.7% low. Reaching the exact $-Z^2/2$ needs $\sigma = 0.771$, 0.932, 1.092 for $Z = 2, 3, 4$, and at those values the kinetic energy is still 11, 19 and 27% low: the total comes right only by cancellation (box $12\,a_0$; a single atom's energy moves by $0.2\,\mathrm{m}E_h$ from there to the declared box). Deeper kernels have sharper cusps, so the softening is the dominant error in every atom heavier than hydrogen, and it is a resolution error rather than a defect of the construction.

Scope of what is reported. The molecular results in Section 8 were computed with the *valence-only* reduction of this model—a kernel of reduced charge Z_x carrying the bonding shell alone, with the core absorbed into the charge reduction. That reduction is cheaper and it is what the tabulated numbers use, but it is not the model above: with the core electrons absent, nothing performs the exclusion, and the valence collapses to $\langle r \rangle = 0.45\,a_0$ against a physical 1.6, which over-binds carbon by a factor of 2.8 against the sum of its ionisation energies. The shell model is stated here because it is what removes that defect at its origin; its own molecular numbers are not yet measured, and nothing in Section 8 should be read as testing it.

5 Hydrogen

We work hydrogen through on the three observables the simulator reports: the binding energy, the geometry the model relaxes to, and the force residual at the observed geometry. H_2 is the natural first case because, as Section 4 notes, its atomic reference carries no pseudokernel, so every error below belongs to the discretisation and the interface treatment rather than to the core model.

Everything below is on the declared grid (5). Before the results, what the box contributes at fixed spacing, because it turns out to contribute more than a grid choice should:

box (a_0)	E(H)	$E(H_2)$	E_{bind}	vs D_e	R
10	−0.486810	−1.147250	0.173630	−0.5%	1.7951
12	−0.486509	−1.145757	0.172739	−1.0%	1.7963
14	−0.486591	−1.143770	0.170588	−2.2%	1.7973
20 (declared)	**−0.486591**	**−1.139790**	**0.166608**	**−4.5%**	**1.7983**

The atomic reference is box-free, varying by $0.0003\,E_h$ across the range, and the bond is box-free, varying by 0.18%. The whole effect is in the molecular energy, which is the interior interface again: the molecule has one and the atom does not. The binding energy spans 4.0 points of D_e, twice the initialisation scatter measured below, so the dependence is real.

It cannot be extrapolated away. The shift per unit box is 0.00075, 0.00099, $0.00066\,E_h$ across the three intervals—not monotone, so not a power law in 1/box—and every individual rung is at or below the scatter even though the overall range is twice it. Neither a convergent limit nor an unbounded expansion of the free boundary is established, and we quote no box-limit value.

Two consequences follow, and both are more awkward than the grid declaration was meant to be. The box joins h and the five coefficients of (6) in the declared set: it is not a convenience sized to hold the molecules but a parameter worth four points of the answer. And the dependence runs the unflattering way—the *smallest* box binds most strongly, within half a percent of D_e, and enlarging it to the declared value costs four points. An earlier figure of six points between boxes 12 and 15 was obtained at fixed *cell count*, so the spacing changed with the box; it is confounded and withdrawn.

The experimental targets are $R_e = 1.40028\,a_0$ and $k = 5.75\,\mathrm{mdyn/Å} = 0.3692\,E_h/a_0^2$. For the dissociation energy there are two standard values and the difference between them exceeds the model's own scatter, so we give both:

		E_h	eV	
D_0	measured	0.16457	4.478	the directly observed value, from the vibrational ground state
D_e	well depth	0.17449	4.748	D_0 plus a zero-point energy of $0.00993\,E_h$

D_e is the matched comparison, because the model carries no nuclear motion: what it computes is the depth of a well, not the energy of a vibrational state. But D_e is a hybrid of a measurement and a spectroscopic correction, whereas D_0 is purely measured, and the two differ by 6.0 percentage points. We quote against D_e and state the D_0 figure alongside it throughout rather than choose silently.

5.1 Binding energy

Table 1 shows that the binding energy is decided by how the interface cells are counted. Where two domains meet, the partition weight w passes through intermediate values, and the kinetic energy of those cells is an artefact of the diffuse representation of a boundary that the model treats as

interface kinetic treatment	$E(\mathrm{H})$	$E(\mathrm{H_2})$	E_{bind}	vs D_e	vs D_0
cut-off at $w = 0.9$, box 20	**−0.486591**	**−1.139790**	**0.166608**	**−4.5%**	**+1.2%**
cut-off at $w = 0.9$, box 12	−0.486509	−1.145757	0.172739	−1.0%	+5.0%
full w-weighting, box 12	−0.486422	−1.134067	0.161223	−7.6%	−2.0%
exterior cleared, box 12	−0.485361	−0.935158	−0.035564	unbound	unbound

Table 1: Binding energy of $\mathrm{H_2}$ under three treatments of the partially occupied cells at the interface, with molecule and atom relaxed for the same number of steps. Note that the two conventions disagree about which of the first two rows is better, the cut-off being 1.0% low against D_e and 5.0% high against D_0 while the uncorrected weighting is 7.6% low and 2.0% low respectively—another reason to quote both. The reference and the molecule must be run to the same step count: an earlier value of −11.4% for the middle row came from an atomic reference stopped at 70 500 steps against a molecule at 232 500, and is withdrawn.

sharp. Excluding cells below an occupancy of 0.9 removes $0.0117\,E_h$ from the molecule and only $0.000087\,E_h$ from the atom—a ratio of 67 to 1 per domain—because the molecule has an interior interface where two domains meet while the atom has only an outer edge with nothing on the far side. The exclusion therefore does not cancel in the difference, and it is the single change that brings the binding energy to experiment.

The third row is the control. Clearing the exterior amplitude altogether, rather than merely declining to count its kinetic energy, unbinds the molecule: E_{bind} goes negative and the relaxed separation runs out to $3.56\,a_0$. The amplitude just outside a domain is what makes the interface flux small, and removing it imposes a drain at the boundary instead.

The uncertainty is measured, not estimated. The exterior amplitude in `mol_fast` is not varied by the relaxation—where w vanishes the update is identically zero—so its value is inherited from the initial state, and the binding energy could in principle be an artefact of that choice. Varying the initial decay constant by a factor of 2.1 gives

z_{init}	0.7	1.0	1.5
E_{bind}	0.175248	0.172739	0.176339
vs D_e	+0.45%	−0.99%	+1.07%
R (relaxed)	1.7993	1.7963	1.7953

a spread of 2.1 percentage points which is not monotone in the seed, so it is scatter rather than a dependence, and a geometry spread of 0.22%. We therefore quote

$$E_{\mathrm{bind}}(\mathrm{H_2}) \;=\; 0.1666{\pm}0.0018\,E_h \;=\; -4.5{\pm}1.0\%\ \text{of}\ D_e \;=\; +1.2{\pm}1.1\%\ \text{of}\ D_0 \qquad \text{[declared box; +4.0 points at box} \tag{8}$$

the central value on the declared grid, the stated uncertainty the measured initialisation scatter, and the bracketed figure the declared-box sensitivity—which is twice the scatter and must be quoted with it. The result lies *between* the two experimental conventions, $D_0 = 0.16456 < 0.16661 < D_e = 0.17447$, closer to the directly measured D_0 than to the well depth it should properly be compared with. So the model's binding is accurate to a few percent on either reading, and which few percent depends on two things it cannot itself resolve: a convention, and a declared box. For scale, the choice of interface condition discussed in Section 5.2 moves the same quantity by 29 to 58 points, so the initialisation is a minor term.

5.2 The interface condition, and why hydrogen cannot decide it

What is actually new, relative to one electron. Section 1 argued that at one electron this model is the minimal Coulomb model and coincides with the standard one. Everything the construction adds is therefore contained in two things: *non-overlapping* one-electron densities, and the condition under which they meet. Nothing else about the one-electron block changes—the kinetic term and the Coulomb attraction are the same. It is worth being explicit that the addition fixes what keeps electrons apart: there is no joint wavefunction here to antisymmetrise, so the only repulsion between two electrons beyond the direct Coulomb term is the fact that they cannot occupy the same space. That subtraction is what the over-binding reported in Section 1 measures, and it is most visible exactly where territorial exclusion can do nothing—between *equivalent* domains facing each other across a mirror plane, where the two constructions below agree to 4×10^{-8} and neither supplies any repulsion at all.

The two conditions are not rivals. They enter with different roles, and the count of equations is what makes this work. For a *fixed* partition, free variation of each density gives homogeneous Neumann on that domain's own boundary—$\partial u/\partial n = 0$ is the natural condition and nothing is imposed. Solved that way the amplitudes at the interface generically *disagree*, and the interface is then placed where they agree. So continuity is not a second assumption about Ψ: it is the *equation that determines* Γ. Three conditions, one extra unknown. The implementations in this programme each take one half of that scheme and supply the other half by hand: `mol_fast` drives the amplitude mismatch to zero and so genuinely *finds* Γ, while imposing the Neumann condition only weakly through $\frac{1}{2}\int w|\nabla u|^2$; the sharp constructions impose Neumann exactly but are *told* where Γ is, by symmetry. Neither is a different theory; both are incomplete implementations of one, and the gap between them is measured below.

The free boundary between two domains needs a condition, and the construction admits two that are both defensible. The model requires $\Psi \in H^1$, that is continuity of the densities across the interface, whose natural condition is continuity of the amplitude, $u_1 = u_2$, with the normal derivative free to jump. `mol_fast` implements this: its front advances on the local amplitude mismatch and converges where that mismatch vanishes. The alternative treats the domains as disjoint with u unconstrained at its own boundary, for which the natural condition is $\partial u/\partial n = 0$; a mirrored ghost value, or equivalently omitting the face from the stencil, imposes it exactly, and two independent implementations in this programme do so.

Omitting the face imposes it exactly only on a *sharp* partition. The qualification matters, because the obvious way to get the strict condition inside this solver does not work. Binarising the face weight—counting a face only where both its cells are owned—while leaving the occupancy diffuse gives H_2 $E = -1.338705$ against -1.127903, a binding energy of $+110\%$, and a dipole of 0.108–0.187 D where reflection symmetry requires identically zero. The cause is diagnosable rather than mysterious. The kinetic energy of the assembled field, $T_\Psi = \frac{1}{2}\int |\nabla \sum_m w_m u_m|^2$, charges the interface *jump* $(u_2 - u_1)^2/h$, which closed faces make free for the per-domain sum; measured, $T_\Psi/T = 8.07$ at $D = 2.1\,a_0$ and 4.25 at $1.3\,a_0$. A factor of eight means the discrete solution is not in H^1: an amplitude discontinuity has opened at the interface and the minimiser is exploiting it, and the per-domain T falls 39% precisely because the gradient is no longer charged where it is largest. We note this is *not* inferred from the energy lying below the exact value, which is not evidence of anything here—the functional is not Schrödinger's and no variational theorem connects its minimum to the exact ground state. So the strict condition is available in this programme only where the partition is sharp *and* its location is supplied by symmetry, which is the same restriction

discussed below.

The two give materially different energies—+27.9% and +58.0% on E_{bind} for the strict condition against +0.2% for the continuity condition, with bond errors of +15.9% and −2.3% against +52%—so the choice is not a detail. But H_2 cannot adjudicate it. A symmetric diatomic has $u_1 = u_2$ at the midplane *and*, by reflection, $\partial u/\partial n = 0$; the two conditions coincide, and the spread above is a difference between discretisations rather than between physical conditions. Settling it requires a two-domain system with no symmetry plane, which is not in the present list.

6 The solver

Two solvers, and the one used here is the less faithful. `molecule.js` carries *one* amplitude field with every cell labelled by its owner, so the domains are non-overlapping *by construction*; it admits two thousand domains and has been run on alpha-helices. `mol_fast`, which produced every number in this paper, carries one array per domain, all overlapping, with the weight w gating only the evolution. One labelled field is the formulation of Section 2; `mol_fast` relaxes it for speed. It also cannot relocate a domain: u evolves only where $w > 0$ and w advances only where u already wins, so neither field can lead and a domain creeps at its boundary without crossing a gap—measured on lithium, where seeds at 0.15, 1.0 and 4.0 a_0 ended at 0.118, 0.799 and 3.203.

This matters least where it might be expected to matter most. The one system for which both treatments are converged is H_2, and they *bracket* the exact energy from opposite sides:

	$E(H_2)$	error	E_{bind} vs D_e
overlapping domains (`mol_fast`)	−1.1398	+3.0%	−4.5%
hard non-overlap (axisymmetric)	−1.2760	−8.6%	+58%
exact	−1.1745	—	—

Hard non-overlap does not recover the exact energy; it overshoots past it, and over-binds by a factor of thirteen more than the overlapping scheme on the binding energy. So *faithful* and *accurate* are different claims here, and only the first is established for `molecule.js`. The honest statement of this paper's energy result is therefore a bracket rather than a value: **H_2's binding lies between −4.5% and +58% of D_e according to how non-overlap is enforced**, and an intermediate choice reproducing experiment would be a fit, not a prediction.

What the comparison does establish is *which observables depend on the approximation*. The energies differ by eleven points between the two schemes; the converged bond lengths do not—1.31 extrapolated from the overlapping scheme against 1.368 from the hard one, both a few percent short. **Geometry is robust against the non-overlap treatment where energy is not**, which is an argument for the geometry-first ordering of Section 1 reached from an independent direction.

The measurements above were made with a GPU solver (`mol_fast`) that carries one domain per entry and caps at sixteen domains. That cap is the binding constraint on molecular size, and its meaning depends on the construction: with one domain per atom carrying Z_x electrons it permits sixteen atoms, which covers benzene, glycine, ethanol and comparable small organics; with the one-electron domains of the underlying construction it permits sixteen electrons, which covers CH_4 and C_2H_2 but not benzene.

The scope of the pseudo-kernel core model itself, measured on two-centre systems, is reported separately [3].

Two properties of `mol_fast` are relevant to reading the numbers above, and we state them rather than leave them implicit. The partition weights are diffuse: w passes through intermediate values over a few cells at an interface instead of switching between 0 and 1, which is what makes the

interface kinetic energy of Section 5.1 a quantity that must be decided rather than a consequence of the model. And the amplitude outside a domain is not relaxed—where w vanishes the update is identically zero—so it is inherited from the initial state. The measured consequence of that is the $\pm 1.0\%$ band on the binding energy, which is small, and bounds how much of the result can be attributed to initialisation rather than to the model.

A second implementation (`molecule.js`) carries the same model as one global field partitioned by labels, enforcing true non-overlap with sharp interfaces, and admits two thousand domains; it has been run on alpha-helices and on chignolin. It is not used for any number reported here. Its interest is that its sharp partition removes the interface ambiguity above at the cost of a different force treatment, so it is the natural setting for the asymmetric two-domain test that Section 5.2 identifies as outstanding.

7 Three checks, and which one applies

Every calculation carries three internal checks. They are not interchangeable, and the choice between them is dictated by symmetry—a point we found easy to get wrong.

Kernel–kernel energy. $V_{KK} = \sum_{a<b} Z_a Z_b / R_{ab}$ is exact arithmetic and confirms the geometry *actually built*, independently of any physics. Every configuration reported below matched its predicted value; this is how fractional kernel placement, per-bond spacing and coincident kernels were verified.

Total force as a convergence criterion. The sum of nuclear forces is *not* automatically zero. It equals the net force the electron cloud exerts on the nuclei; only when the electronic state is stationary does the cloud exert no net force, and then $\sum_a \mathbf{F}_a = 0$ follows by Newton's third law. The residual is therefore a direct measure of electronic non-convergence—and a more reliable one than any energy-based criterion, since a relaxation can sit on a long plateau with the energy changing imperceptibly while the state is far from settled.

But symmetry cancels it. In a symmetric molecule, mirror or rotational symmetry forces $\sum_a \mathbf{F}_a = 0$ regardless of accuracy. A small residual in CO_2 (we measure 0.06% of the largest force) establishes only that the algorithm preserves symmetry. The sum carries information only in directions symmetry does not protect: every direction for a heteronuclear molecule, the C_3 axis of NH_3, the C_2 axis of water.

Per-atom zeros. Where symmetry is complete, the useful check is instead a single atom's force. An atom at a centre of symmetry—the carbon of CO_2 or of CH_4—must feel exactly nothing. Unlike the sum, this cannot be satisfied by cancellation among atoms, so a nonzero value is unambiguously error and its magnitude is the per-atom floor.

8 Results: relaxed geometries and binding energies, with forces as a check

Table 2 is the paper's primary table: the two predicted quantities with the dipole beside them. Table 3, which follows, is the force *check* and the older binding-energy set, kept because it is the only coverage of the polyatomic rows.

The ten remaining rows are being measured on this grid. Until they are, the binding energies in Table 3 stand as the only polyatomic coverage, and seven of its nine populated entries come from the superseded grid—annotated there, but old.

Table 3 collects the configurations. Forces are quoted with + meaning outward, so a positive entry indicates the bond wants to lengthen. Where a configuration fails its own check we report the failure rather than a bond error.

Two columns of this table require the qualifications established on hydrogen, and we give them before the table rather than after it. The ΔR entries were converted with the *measured* force constant, so each is a *lower bound* on the model's bond error, too small by the ratio of the measured curvature to the model's own; for H_2 that ratio is 1.97 and the entry below should read +52% rather than +30%. Correcting the remaining rows requires a three-point scan per configuration to measure each model curvature, which has not been done. And the last column is a diagnostic, not a score: an H_2 configuration that does not bind at all returns 2.1% on it, which would place it among the best entries here. Its only legitimate reading is the symmetry-dependent one given in the caption.

Only the declared valences appear. Every row is at $Z_x = 1, 4, 3, 3$. The C+2 and O+2 variants are not results of the model but the alternatives it rejects, measured as such in separate convergence and valence studies, and carrying them here would present a ruled-out convention as a finding. Removing them has one consequence worth stating plainly: **CO and CO_2 leave the table altogether**, because the only force calculations ever made for them were at O+2. There is no force measurement for either molecule at the valences the simulator actually uses, and the earlier reading of CO_2's -7.2% as an ordinary bounded error was therefore a statement about a configuration this paper rejects. What is measured for CO_2 at $3, 4, 3$ is its binding energy, and it does not bind (the across-composition comparison).

Precision of the binding-energy column. The six decimals are the solver's, not the measurement's, and most of this column carries no established precision at all. Its error is dominated by a *systematic*: the one row measured on both grids moves by $0.0555\,E_h$, or 32 points of D_e (H_2: 0.1111 against 0.1666), which is larger than several of the entries themselves—N_2 at 0.0155, CO at 0.0137, O_2 at 0.0770. Mixing boxes between rows contributes a further $0.023\,E_h$ (CO 0.0369 against 0.0137 matched), the box at fixed h some 0.007, and the step count a similar amount, these rows having been stopped at 119 500 steps against the 232 500 used elsewhere. No *random* component was ever measured on them: no seed variation and no box ladder. The operational consequence is a single rule—**an entry below about $0.056\,E_h$ cannot be distinguished from zero on this evidence**—and N_2 is the demonstration, its -0.0155 becoming $+0.2764$ once the grid was matched. Only two rows carry a stated uncertainty: H_2 at 0.1666 ± 0.0018 from measured initialisation scatter, with a further four points of declared-box sensitivity; and N_2 at +0.2764 with a 6% reference-shift uncertainty, which secures the sign and not the value.

The binding-energy column carries a third and sharper qualification. Only hydrogen is on the declared grid; the rest predate it, and the single row available on both grids shifts by 35 points of D_e between them. That shift is larger than every deficit in the column, so no entry here establishes that a molecule fails to bind—a statement we had drawn from this set and now withdraw. What the column does establish is the direction and rough scale of the construction's binding on a common grid, and that the earlier grid understated it severely.

8.1 H_4 against an exact reference

One configuration can be scored against an exact method rather than experiment. Full configuration interaction in a minimal basis [9] places the equilibrium of the equally spaced H_4 chain at $a = 1.8\,a_0$, and the chain has one electron per site with no pseudopotential, so no core parameter is involved. At that geometry all four RealQM forces point outward—the model would expand a chain the exact method calls stationary—while over-binding by 14–18% per atom when each method is referred to its own atom. Too long is the pattern H_2 shows at +52% on H_2. The over-binding is less clear-cut than it reads: the reference is a full configuration-interaction calculation in a minimal STO-6G basis, whose cohesive energy is itself 14% low against reality (0.0766 against $0.0872\,E_h$ per atom), so over-binding against it is not the same statement as over-binding against experiment.

9 Limitations

The bond errors reported here are not competitive with established electronic structure methods, and are not offered as such; they measure where this real-space construction currently stands. Several specific limitations bound the numbers.

The construction is not parameter-free. Besides the declared spacing it carries the five coefficients of the free-boundary update listed in Section 3, four of which have measured sensitivities below three points on H_2 and one of which (μ, the front speed) has never been laddered on a molecule. It also takes the valence charges Z_x from separate atomic work, and the kinetic occupancy cut-off at 0.9 is a declared choice worth 6.6 points of binding energy. None of these is fitted to the quantities reported here, but none is derived either.

All calculations use bare kernels, so the deepest, $Z = 4$, sits at $Zh = 0.48$, far above the resolution guideline. Mesh ladders are deliberately *not* used to justify the numbers, for the reason given in Section 3; the one ladder we do report, for H_2's bond, shows the refinement sequence passing through experiment rather than converging on it, which is a limitation of the construction and not of the sampling.

The convergence problem is characterised but not solved. We can now measure electronic non-convergence, attribute it principally to shallow kernels, and exclude three rival explanations; we cannot yet remove it. Configurations containing shallow kernels or threefold and bent geometries remain unusable, which includes water and ammonia—a serious limitation for any application to hydrogen-bonded systems.

Measured geometries, force constants and dissociation energies are taken from the standard compilations [7, 8] and are recorded as unverified in the source files. Dissociation energies carry the further ambiguity of Section 5.1: D_0 is measured and D_e adds a spectroscopic zero-point correction, the two differ by six points for H_2, and only D_e is the quantity a model without nuclear motion computes.

Finally, the construction pins the electron count of each domain, so charge transfer between unlike atoms is representable only through the shape of each domain and not through its occupation. For homonuclear systems this costs nothing; for a heteronuclear molecule it constrains precisely the asymmetry that defines it.

10 Conclusion

The simulator is defined by a declared grid rather than by a convergence limit: one spacing for every molecule, with the atomic references that enter the binding energy becoming constants of the

model rather than per-run bookkeeping. That choice gives up the description of the construction as parameter-free, and we have listed what it actually carries. What it gains is that each number is answerable: with the grid fixed there is no remaining adjustment to defer a disagreement to.

Hydrogen, the one system whose atomic reference involves no pseudokernel, comes out as a single defect rather than three. Its depth is right to a few percent—$0.1666 \pm 0.0018\,E_h$, falling between the two experimental conventions, with the uncertainty measured over a factor of two in the initialisation rather than estimated. Its gradient is right, the volume force predicting the model's own minimum to 0.2% from one calculation at the observed geometry. Its curvature is 49% too soft, and its bond is 52% too long *because* of that, through $\Delta R = F/k_{\mathrm{model}}$. One quantity is wrong and the other errors follow from it, which is a more useful finding than three independent discrepancies would have been, and it identifies the curvature as what any improvement has to address.

And nitrogen binds. Measured with molecule and reference on one grid, $E_{\mathrm{bind}} = +0.2764\,E_h$ in a symmetric state, where the same quantity computed across two grids gave -0.0155 and was reported as a failure to bind the strongest common bond. That reversal is the single most consequential result here, and it is a lesson about method rather than about nitrogen: a number measured on one grid is not evidence about another. The same comparison moves hydrogen's binding by thirty-two points and carbon's energy by $1.1\,E_h$, and it is why the binding column of Table 3 carries a systematic larger than several of its own entries—nothing below about $0.056\,E_h$ in it can be distinguished from zero.

Three further things bear on how the earlier protocol should be read. The residual total force is a convergence diagnostic and not a figure of merit: an unbound H_2 scores as well on it as the best entries in the table. Bond errors converted with the measured force constant are lower bounds, too small by the stiffness ratio wherever the model's well is soft—twenty-five points for H_2. And the long bond is a *resolution* artefact, not a property of the construction: an axisymmetric reduction, exact for two domains on an axis, converges the same bond to $1.368\,a_0$ and holds it flat to 0.18% under a two-and-a-half-fold refinement, so the model's own bond error is -2.3% where the declared grid reports $+52\%$.

What remains outstanding is specific rather than general. The front-speed coefficient has never been laddered on a molecule. The choice of interface condition, worth 29 to 58 points of binding energy, cannot be settled on any symmetric system—H_2 included—because symmetry makes the candidates coincide. The atomic references are the binding energies' real bottleneck, the valence-only reduction over-binding carbon by a factor of 2.8; the shell construction of Section 4 is stated as what removes that at its origin, and its molecular numbers are not yet measured. And the two conventions for a dissociation energy differ by more than the model's own scatter, so every binding figure here is quoted against both.

References

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[2] C. Johnson, *RealQM for Atoms and the Periodic Table*. The shell configurations used here, including oxygen as 2+3+3, are obtained there from atomic calculations independent of any molecular input.

[3] C. Johnson, *The Two-Centre Test: where the pseudo-kernel core model works, and where it stops*. Companion study measuring three core models—hard depletion, clamp and empty core—against the alkali and halogen dimers.

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[9] Full configuration interaction in a minimal (STO-6G) basis for equally spaced hydrogen chains H_N, generated for this work: equilibrium at $a = 1.8\,a_0$ with $E = -2.19038$ ($N = 4$) and -3.26674 ($N = 6$), each referred to its own atom $E(\mathrm{H}) = -0.471039$. The hydrogen chain is a standard benchmark for many-electron methods, having one electron per site and no pseudopotential, so no core parameter enters the comparison.

[10] Public code, the molecule simulator of the simulator, and an interactive gallery of all experiments in the present work: `https://claes542.github.io/RealMolecule/gallery.html` (source repository: `https://github.com/Claes542/RealMolecule`).

system	Z_x	E_{bind}	μ (D)	μ_{exp}	$\lvert\sum F\rvert$	state
N_2	3,3	0.2131	0.038	0	1.2%	accepted
O_2	3,3	0.2156	0.012	0	5.4%	accepted
CO	4,3	0.2219	0.154	0.110	6.3%	accepted
CO_2	4,3,3	0.3440	0.000	0	0.1%	accepted
N_2O	3,3,3	0.3915	0.023	0.161	7.4%	accepted
CH_4	4,1,1,1,1	1.0823	0.004	0	3.4%	accepted
HCN	1,4,3	*(0.4371)*	*(0.165)*	2.985	83%	rejected
H_2O	3,1,1	*(0.5164)*	*(0.938)*	1.855	117%	rejected
NH_3	3,1,1,1	*(0.6834)*	*(0.786)*	1.472	127%	rejected

Table 2: **These rows predate the calibration** (1) **and the declared box, and are being re-measured; they are retained here only so the structure of the table is visible.** They were computed at $\sigma = 1$, against (5)'s calibrated $\sigma = 0.627$; the box and spacing are already those of (5). The difference is material, since σ sets the cusp resolution and hence every energy. Primary results, one grid throughout (box $16\,a_0$, $h = 0.10$, $r_c = 0$, occupancy cut-off 0.9, 232 500 steps) against atomic references measured identically: $E(\mathrm{H}) = -0.486591$, $E(\mathrm{C}) = -15.113441$, $E(\mathrm{N}) = E(\mathrm{O}) = -7.252538$. **The state column is an acceptance test.** Where symmetry requires $\mu = 0$ (N_2, O_2, CO_2, CH_4) the computed value is a pure error measurement needing no reference, and all four pass—0.000 to 0.038 D. Where no symmetry protects the total force (CO, N_2O, HCN) the residual is a genuine convergence measurement and admits rows below 20%. Rejected rows are shown parenthesised so the rejection can be checked, never as results; three of the four have an *unprotected direction*—H_2O and NH_3 a C_2 or C_3 axis, HCN no symmetry at all—and settle to a *stable* asymmetric state rather than converging slowly: water's residual held at 116.7, 116.5, 116.6% across its run with its two hydrogens equivalent to 0.1%, so the imbalance lies along the axis and more steps will not remove it. **Hydrogen is deliberately absent.** Its repeat run on this grid did not converge, while it converges at boxes 12 and 20—so that is a bad grid point, not a failed molecule, and a different thing entirely from the rejections above, which are *stable* wrong states. Listing it here would conflate the two. Hydrogen is treated in full in Section 5, where its binding is -4.5% of D_e, its bond converges to $1.368\,a_0$, and its force agrees with $-dE/dR$ to 0.06%. R_{min} is omitted: it exists for H_2 alone, no commensurate scan having been run for the others. The two polar accepted rows give +40% (CO) and −86% (N_2O) against measurement—CO's dipole being famously small at 0.11 D—so the dipole magnitudes are poor where the null tests are excellent.

system	Z_x	force	ΔR from F/k_{exp} ∗	$\lvert\sum F\rvert/\lvert F\rvert_{\max}$	E_{bind} §
H_2	1,1	+0.1383	+30% ∗	14.7% †	**+0.1666**
N_2	3,3	−0.2505	−8.2%	3.7% †	**+0.2764 ¶**
O_2	3,3	−0.2463	−14.3%	8.4% †	+0.0770
HCN	**1,4,3**	C≡N −7.8%, H–C +3.5%		**5.5**%	+0.6099
H_2O	3,1,1	—		126% ‡	+0.4307
H_3 lin.	1,1,1	+0.0420	—	34.9% †	—
H_4 chain	1,1,1,1	+0.0316	—	7.4–16% †	—
C_2H_2	1,4,4,1	outward	—	25% †	+0.3177
C_2N_2	3,4,4,3	N inward, C outward		3.2% †	+1.9819
NH_3	3,1,1,1	—		141% ‡	+0.5424
NH_3 planar	3,1,1,1	—		70% ‡	—
CH_4	4,1,1,1,1	0.036 outward	—	0.6% †	+0.4081

Table 3: The force check at measured geometries, with binding energies. **Neither of the paper's two primary observables appears in full here**: the model's own relaxed geometry is measured only for H_2 (+52% from a commensurate scan at box 20; a scan at box 12 gives +28%, so the figure carries a box dependence of its own), and the ΔR column is *not* that quantity—it is the force residual divided by the *experimental* force constant, which understates the model's error by twenty-five points on the one row where both are known. The $|\sum F|$ column is a convergence diagnostic to which no verdict attaches, because what it *measures* depends on symmetry. †: every direction is symmetry-cancelled, so the entry reports whether the algorithm preserves symmetry and bounds nothing about convergence. ‡: cancelled except along the unique axis, so the residual is a convergence measurement in that direction. Unmarked rows (CO, HCN) have no symmetry, and their entries are genuine convergence measurements—note that these are among the largest in the table. Rows where no ΔR is given are those whose own checks exceed the signal. ∗: converted with the *experimental* force constant, hence a lower bound on the model's bond error. For H_2 the measured minimum is +52% against this column's +30% on H_2, and the same correction applies in unknown measure to every other row, none of which has had its geometry scanned. §: only the H_2 entry (bold) is on the declared grid of (5) with the $w = 0.9$ cut-off; every other binding energy comes from an earlier set at the default valences in a box of $15\,a_0$ with $h = 0.1389$, $\varepsilon = 0.14$ and the uncorrected kinetic accounting. ¶: H_2 is the one row measured both ways, and it moves by $+0.0555\,E_h$, or 32 points of D_e, between them—6.6 of those points from the cut-off and the rest from the grid and softening. Every deficit in this column is smaller than that shift, so *the unbound entries are not results*. That has now been measured rather than suspected: re-run with molecule and reference on one grid (box 10, $h = 0.10$), **N_2 binds by** $+0.2764\,E_h$, from $E(\mathrm{N}) = -7.260914$ and $E(N_2) = -14.798207$, with a symmetric state ($\Sigma F = 5.5\%$). The atomic reference shifts by only $0.008\,E_h$ between boxes 10 and 16, i.e. 6% of the binding energy, so the sign is secure. The earlier -0.0155 came from box 15 with $h = 0.1389$ and is withdrawn. Blank entries are rows whose valences or geometry were not in the earlier set; a row is deliberately left empty rather than filled from a run at different Z_x. There is no experimental column because the repository holds no verified table of atomisation energies, and H_2 is the only row with a reference we have checked. Configurations in which two different elements receive the same Z_x are degenerate—the model cannot distinguish them—and are omitted.